



Aror University of Art, Architecture, Design & Heritage Sukkur.

BS AI/MMG/CYS 2026

Data Structures and Algorithms

Course Title:	Data Structures and Algorithms
Course Code:	CSC-221
Credit Hours:	(3)
Course Instructor:	Faria Naz
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Description:

Develop a strong foundation in efficient data organization and algorithm design through this comprehensive Data Structures and Algorithm Analysis course designed for undergraduate students. The course introduces fundamental concepts such as abstract data types (ADTs), algorithm analysis, asymptotic notation, recursion, and divide-and-conquer techniques. Students will explore both linear and non-linear data structures, including arrays, linked lists, stacks, queues, trees, heaps, hash tables, and graphs, along with a wide range of searching and sorting algorithms. Emphasis is placed on analyzing algorithm efficiency using time and space complexity, selecting appropriate data structures for different computational problems, and implementing optimized solutions. Through programming assignments, practical exercises, and real-world problem-solving activities, students will develop the analytical and programming skills required to design efficient software systems and prepare for advanced courses in computer science and software engineering.

Aims and Objectives / Description:

This course introduces the fundamental concepts of Data Structures and Algorithm Analysis, emphasizing efficient data organization, algorithm design, and complexity analysis. Students will learn to implement and analyze linear and non-linear data structures, apply searching and sorting techniques, and evaluate algorithm performance using time and space complexity. Through theoretical concepts and practical programming exercises, the course develops problem-solving skills essential for designing efficient software solutions and advanced computing applications.



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Marks Distribution

S. No	Assessment Activities	Percentage	Total Activities
1.	Sessional: Quizzes/ Assignments (Quizzes & Assignments)	30%	1
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Course Learning Outcomes (CLOs):

No.	Course Learning Outcome	Domain	Level	Assessment Tool
C1	Explain the fundamental concepts of abstract data types, algorithm analysis, asymptotic notation, and linear data structures.	C	2	Class Participation, Quizzes, Mid Exam, Assignments
C2	Implement and analyze linear and non-linear data structures and apply appropriate searching and sorting algorithms to solve computational problems.	C	3	Class Activities, Quizzes, Assignments, Lab Exercises
C3	Design efficient solutions using advanced data structures and graph algorithms, and evaluate algorithm performance based on time and space complexity.	C	4	Worksheets, Programming Project, Presentations, Final Examination

Bloom's Taxonomy Alignment:

CLO	Bloom's Level	Justification
C1	C2 – Understanding	Students explain ADTs, complexity analysis, recursion, and the properties of data structures.
C2	C3 – Applying	Students implement stacks, queues, linked lists, trees, hashing, and sorting/searching algorithms in programming assignments.
C3	C4 – Analyzing	Students compare algorithms, analyze time and space complexity, and select suitable data structures for solving real-world problems.



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Domains:

C=Cognitive, A=Affective, P=Psychomotor

Levels:

Cognitive = {1: Remembering, 2: Understanding, 3: Applying, 4: Analyzing, 5: Evaluating, 5: Creating}

Affective = {1: Receiving, 2: Responding, 3: Valuing, 4: Organizing, 5: Characterizing}

Psychomotor= {1: Imitation, 2: Manipulation, 3: Precision, 4: Articulation, 5: Naturalization}

Text Books

1. Data Structures By Seymour Lipschutz

Reference Books

1. Data Structures with C++ by John R. Hubbard
2. Data Structures and Algorithms in Java by Dan S. Myers
3. Data Structures and Algorithms for BE Anna University

Detailed Course Outline:

Weeks	Lec#	SUBTOPICS	Reference	Course % Covered
Week No. 01	Lec:01	Introduction to Data Structures • Data Structures Overview• Primitive & Non-Primitive Data Types• Abstract Data Types (ADT)	Ch-1	2.08%
	Lec:02	Algorithm Analysis • Characteristics of Algorithms• Time & Space Complexity	Ch-1	4.16%
	Lec:03	Asymptotic Analysis • Big-O• Big-Ω• Big-Θ Notation	Ch-1	6.25%
Week No. 02	Lec:04	Recursion • Recursive Functions• Recursive Thinking	Ch-2	8.33%
	Lec:05	Analyzing Recursive Algorithms • Recurrence Relations• Recursive Complexity	Ch-2	10.41%
	Lec:06	Divide and Conquer • Concept• Binary Search• Merge Strategy	Ch-2	12.50%
Week No. 03	Lec:07	Arrays & Searching • Linear Search• Searching Unsorted Arrays	Ch-3	14.58%
	Lec:08	Binary Search • Sorted Arrays• Complexity Analysis	Ch-3	16.66%



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	Lec:09	Linked Lists • Singly Linked List• Node Representation	Ch-4	18.75%
Week No. 04	Lec:10	Linked Lists • Doubly Linked List• Circular Linked List	Ch-4	20.83%
	Lec:11	Sorted Linked Lists • Insertion• Deletion• Searching	Ch-4	22.91%
	Lec:12	Applications of Linked Lists	Ch-4	25.00%
Week No. 05	Lec:13	Stacks • ADT Stack• Array Implementation	Ch-5	27.08%
	Lec:14	Stacks • Linked List Implementation	Ch-5	29.16%
	Lec:15	Applications of Stacks • Expression Evaluation• Parenthesis Matching	Ch-5	31.25%
Week No. 06	Lec:16	Queues • Queue ADT• Array Implementation	Ch-6	33.33%
	Lec:17	Queues • Linked List Implementation• Circular Queue	Ch-6	35.41%
	Lec:18	Deque & Priority Queue • Double-ended Queue• Priority Queue Concepts	Ch-6	37.50%
Week No. 07	Lec:19	Sorting Algorithms • Bubble Sort	Ch-7	39.58%
	Lec:20	Sorting Algorithms • Selection Sort	Ch-7	41.66%
	Lec:21	Sorting Algorithms • Insertion Sort	Ch-7	43.75%
Week No. 08	Lec:22	Efficient Sorting • Merge Sort	Ch-7	45.83%
	Lec:23	Efficient Sorting • Quick Sort	Ch-7	47.91%
	Lec:24	Mid Semester Review & Mid Examination	—	50.00%
Week No. 09	Lec:25	Sorting Algorithms • Heap Sort	Ch-7	52.08%
	Lec:26	Sorting Algorithms • Shell Sort	Ch-7	54.16%
	Lec:27	Non-Comparison Sorting • Radix Sort• Bucket Sort	Ch-7	56.25%
Week No. 10	Lec:28	Hashing • Hash Functions• Hash Tables	Ch-8	58.33%
	Lec:29	Collision Resolution • Open Addressing	Ch-8	60.41%
	Lec:30	Collision Resolution • Chaining	Ch-8	62.50%
Week No. 11	Lec:31	Trees • Terminologies• Binary Trees	Ch-9	64.58%
	Lec:32	Tree Traversals • Preorder• In-order• Post-order	Ch-9	66.66%
	Lec:33	Binary Search Trees • Insert• Delete• Search	Ch-9	68.75%
Week No. 12	Lec:34	Heaps • Min Heap• Max Heap	Ch-10	70.83%
	Lec:35	M-Way Trees	Ch-10	72.91%



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	Lec:36	Balanced Trees • AVL Trees• Introduction to Red-Black Trees	Ch-10	75.00%
Week No. 13	Lec:37	Graphs • Graph Terminologies• Representation	Ch-11	77.08%
	Lec:38	Graph Representation • Adjacency Matrix• Adjacency List	Ch-11	79.16%
	Lec:39	Graph Traversals • Breadth First Search (BFS)	Ch-11	81.25%
Week No. 14	Lec:40	Graph Traversals • Depth First Search (DFS)	Ch-11	83.33%
	Lec:41	Topological Sorting	Ch-11	85.41%
	Lec:42	Shortest Path Algorithms • Dijkstra's Algorithm	Ch-11	87.50%
Week No. 15	Lec:43	Advanced Graph Algorithms • Minimum Spanning Tree (Introduction)	Ch-11	89.58%
	Lec:44	Memory Management • Dynamic Memory Allocation	Ch-12	91.66%
	Lec:45	Garbage Collection • Memory Leaks• Automatic Memory Management	Ch-12	93.75%
Week No. 16	Lec:46	Comprehensive Review • Complexity Analysis• Data Structures Comparison	Revision	95.83%
	Lec:47	Problem Solving Session • Integrated Practice	Revision	97.91%
	Lec:48	Final Revision	Revision	100%